LOTQ-0009 / ALGEBRA / BEGINNER / FIELD GUIDE

Algebra Foundations for Quant Research

Variables, equations, constraints, scaling, PnL formulas, and formula debugging.

Library of the Quant

Educational reference manual

Manual Profile

LIBRARY ID	LOTQ-0009
CATEGORY	Algebra
DIFFICULTY	Beginner
FORMAT	Single-column field-guide manual
USE	Study, lookup, and research-system reference

Table of Contents

ALGEBRA / BEGINNER / USAGE

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How to Use This Manual

CONCEPT CARDS - PART 2

READING PATH

Use this manual as a quick recall system, not a textbook. First read the one-sentence meaning and memory jogger. Then check the formula block and market example. Finally, connect the concept to your research system: data, features, model, signal, backtest, risk, or execution.

REPEATABLE CARD STRUCTURE

Card module	How to use it
Meaning	Fast definition in plain English.
Memory jogger	The mental shortcut to recall the concept later.
Quant use	Why the concept matters for research and trading systems.
Formula / worked example	The algebraic form plus a concrete market calculation.
System fit	Where the concept lives in a research architecture.
Confusions	The bug or misconception most likely to break the idea.

ALGEBRA / BEGINNER / CONCEPT MAP

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Algebra Research Map

ONE-SENTENCE MAP

Algebra is the layer that turns market observations into computed features, rules, constraints, sizes, PnL, and validation checks.

PROCESS FLOW: WHERE ALGEBRA ENTERS A QUANT RESEARCH SYSTEM

Raw market data → **Algebraic features** → **Rules and constraints** → **Model or signal** → **Backtest PnL** → **Risk and validation**

The same algebra must be consistent from research notebook to production: formulas, units, signs, denominator checks, and timestamp alignment should not change between layers.

ALGEBRA CHECKPOINTS BY LAYER

Research layer	Algebra responsibility	Typical failure
Data	Units, scale, valid domains	Zeros, missing values, bad symbols
Features	Ratios, logs, sums, normalization	Leakage, wrong denominator
Signals	Inequalities, thresholds, piecewise rules	Boundary bugs, inverted signs
Backtest	PnL, position sizing, recursion	Multiplier or timestamp error
Risk	Weights, constraints, squared penalties	Unbounded leverage, unit mismatch

QUICK RECALL

Recall: the formula is not finished until you know its inputs, output units, valid domain, timestamp rule, and failure mode.

ALGEBRA / BEGINNER / NOTATION

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Notation and Symbol Table

HOW TO READ NOTATION

Subscripts usually identify time or asset. Parameters usually stay fixed during one experiment. Greek letters often represent summary statistics, sensitivities, or model parameters. Always translate notation into plain English before coding it.

COMMON NOTATION USED IN THIS MANUAL

Symbol	Plain-English meaning	Quant example
P_t	Price at time t	Last close, midprice, or bar price.
r_t	Return at time t	Percent or log return for one period.
x_t	Generic observed value	Feature input at time t.
n	Lookback length / sample size	20 bars, 252 trading days.
w_i	Weight on asset i	Portfolio allocation fraction.
μ	Mean / average	Average return or feature value.
σ	Standard deviation	Volatility or feature dispersion.
α	Smoothing or intercept parameter	EMA alpha, model intercept.
β	Sensitivity / slope	Market beta, factor loading.
Σ	Covariance matrix or summation symbol by context	Portfolio risk input; distinguish from sum notation.
ε	Small error or epsilon guard	Tolerance for near-zero denominator.

NOTATION EXAMPLE

```
z_t = (x_t - μ_train) / σ_train

x_t = current feature value
μ_train, σ_train = fitted on training data
z_t = standardized feature at time t

Translation: compare the current value to a training baseline, measured in standard-deviation units.
```

QUICK RECALL

Rule: before using a symbol in code, write the column name, unit, timestamp, and whether it is data or a parameter.

ALGEBRA / BEGINNER / FOUNDATIONS

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Variables, Constants, and Parameters

MEANING AND MEMORY JOGGER

Meaning: A variable changes, a constant stays fixed, and a parameter is a chosen setting that controls a formula or model.
Memory jogger: Variable = live input. Constant = fixed number. Parameter = knob you set before running the system.

DEEPER EXPLANATION

In quant work, most formulas are built from observed variables such as price, volume, return, spread, and volatility. Constants and parameters make the formula operational: a 20-day lookback, a 2.0 risk multiple, or a 0.01 transaction-cost estimate.

WHY IT MATTERS IN QUANT RESEARCH

Separating variables from parameters prevents backtest leakage. The system should learn or choose parameters only on allowed training data, then apply them to unseen data.

SIMPLE MARKET / TRADING EXAMPLE

In SMA_20(t) = average(P_t, ..., P_{t-19}), P_t is the changing price variable and 20 is the lookback parameter.

FORMULA INTUITION

```
signal_t = 1 if P_t > SMA_n(t) else 0
P_t = variable, n = parameter, 1/0 = coded action constants
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Store parameters in config files or experiment metadata. Store variables in timestamped dataframes. Never silently hard-code research knobs inside feature code.
Where it fits in the research system	Data layer supplies variables -> feature layer applies formulas -> experiment layer records parameters -> validation layer checks stability.
Confusions to avoid	Do not call every number a constant. A lookback of 20 is fixed inside one run, but it is still a parameter because you can choose and test it.
Related terms	Expressions, functions, parameters, feature engineering, experiment tracking.

QUICK RECALL

1. Which symbols change every timestamp?
2. Which numbers did you choose before the test?
3. Where should each be stored in code?

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Expressions vs Equations

MEANING AND MEMORY JOGGER

Meaning: An expression produces a value; an equation states that two expressions are equal.
Memory jogger: Expression = compute something. Equation = solve or assert something.

DEEPER EXPLANATION

A formula like P_t / SMA_t is an expression because it returns a ratio. Setting P_t / SMA_t = 1.03 turns it into an equation that asks when price is 3 percent above its moving average.

WHY IT MATTERS IN QUANT RESEARCH

Research code constantly switches between computing features and solving thresholds. Confusing the two makes signals ambiguous and tests hard to debug.

SIMPLE MARKET / TRADING EXAMPLE

Feature: spread_t = bid_t - ask_t. Rule: spread_t <= max_spread. The first creates a value; the second checks a condition.

FORMULA INTUITION

```
expression: x + 2y
equation: x + 2y = 10
condition: x + 2y > 10
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Name computed expressions as columns, then express trading rules as boolean masks. Example: df["above_ma"] = df.price / df.sma > 1.02.
Where it fits in the research system	Expressions live in feature tables. Equations and inequalities live in rule logic, calibration, constraints, and validation checks.
Confusions to avoid	Do not treat a formula output as a trade rule until you define the equality or inequality that turns it into a decision.
Related terms	Equality, inequalities, boolean masks, thresholds, feature columns.

QUICK RECALL

Write one expression and one equation using price, cost, and position size.

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Equality, Equivalence, and Legal Algebra Moves

MEANING AND MEMORY JOGGER

Meaning: Equality means both sides have the same value; legal algebra moves preserve that equality.
Memory jogger: Whatever you do to one side, do to the other side unless you are deliberately changing the problem.

DEEPER EXPLANATION

Algebra lets you rewrite a formula into a more useful form without changing its meaning. This matters when solving for position size, required win rate, break-even cost, or implied price movement.

WHY IT MATTERS IN QUANT RESEARCH

Many trading formulas are rearrangements. If the rearrangement is wrong, the system may size positions, stops, or expected returns incorrectly.

SIMPLE MARKET / TRADING EXAMPLE

If risk = shares * dollars_at_risk_per_share, then shares = risk / dollars_at_risk_per_share.

FORMULA INTUITION

```
a = b
a + c = b + c
a * c = b * c, if c is applied to both sides
a / c = b / c, if c != 0
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Write unit tests for rearranged formulas. Feed known inputs into both original and rearranged forms and verify they agree within tolerance.
Where it fits in the research system	Equation transformations sit inside sizing engines, calibration code, risk checks, and PnL attribution.
Confusions to avoid	Dividing by zero, flipping inequality signs with negatives, and forgetting parentheses are the most common legal-move failures.
Related terms	Rearranging formulas, solving for unknowns, constraints, unit tests.

QUICK RECALL

What algebra move turns risk = shares * stop_distance into shares = risk / stop_distance?

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Order of Operations and Formula Parsing

MEANING AND MEMORY JOGGER

Meaning: Order of operations tells the system which parts of a formula are evaluated first.
Memory jogger: Parentheses beat powers, powers beat multiply/divide, multiply/divide beat add/subtract.

DEEPER EXPLANATION

Humans often read formulas from left to right, but software follows parsing rules. A missing pair of parentheses can completely change a trading feature.

WHY IT MATTERS IN QUANT RESEARCH

Feature definitions must be exact. A ratio, spread, or normalized return can have different values if parentheses are wrong.

SIMPLE MARKET / TRADING EXAMPLE

(P_t - SMA_t) / SMA_t measures percent distance from average. P_t - SMA_t / SMA_t equals P_t - 1, which is usually nonsense.

FORMULA INTUITION

```
correct: distance = (price - sma) / sma
incorrect: distance = price - sma / sma
Python follows the same precedence rules.
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Use explicit parentheses even when they are not strictly required. Format formulas so code reviewers can visually parse numerator and denominator.
Where it fits in the research system	Formula parsing is part of reproducibility. A research engine should expose final feature definitions clearly enough to audit.
Confusions to avoid	Do not rely on memory when mixing subtraction, division, powers, and boolean comparisons. Parenthesize the intended meaning.
Related terms	Expressions, ratios, normalization, feature definitions, code review.

QUICK RECALL

Why is (a - b) / b different from a - b / b?

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Fractions as Scaling Operators

MEANING AND MEMORY JOGGER

Meaning: A fraction is division, but in research it usually acts as a scaling operator that converts raw size into relative size.
Memory jogger: Fractions answer: compared to what?

DEEPER EXPLANATION

Raw price changes are hard to compare across assets. Dividing by a base value converts a move into a relative move, allowing NQ, ES, oil, and a stock to be compared on a common scale.

WHY IT MATTERS IN QUANT RESEARCH

Most useful features are scaled: return = change / price, allocation = position / capital, spread fraction = spread / midprice.

SIMPLE MARKET / TRADING EXAMPLE

A $2 move on a $20 stock is huge. A $2 move on a $500 stock is small. Fractions reveal the difference.

FORMULA INTUITION

```
relative_change = (new_value - old_value) / old_value
allocation = position_value / account_equity
spread_fraction = bid_ask_spread / mid_price
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Guard every denominator. Replace zero denominators with NaN, filter invalid rows, or use a documented fallback. Silent infinities contaminate features.
Where it fits in the research system	Scaling converts raw market data into comparable research features. It is a core step before ranking, modeling, and risk aggregation.
Confusions to avoid	A small denominator can make a fraction explode. Large ratios may mean denominator instability, not a real signal.
Related terms	Ratios, percent change, normalization, denominator checks, outliers.

QUICK RECALL

What denominator makes a raw quantity comparable across assets?

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Ratios and Proportions

MEANING AND MEMORY JOGGER

Meaning: A ratio compares one quantity to another; a proportion states that two ratios are equal.
Memory jogger: Ratio = A compared with B. Proportion = two comparisons match.

DEEPER EXPLANATION

Ratios are everywhere in market research: price to moving average, volume to average volume, risk to reward, win size to loss size, position to equity.

WHY IT MATTERS IN QUANT RESEARCH

Ratios are scale-normalized features. They are often more portable across instruments than raw values.

SIMPLE MARKET / TRADING EXAMPLE

volume_ratio = today volume / 20-day average volume. A value of 2.0 means volume is twice its recent baseline.

FORMULA INTUITION

```
ratio = A / B
proportion: A / B = C / D
solve: A = B * C / D
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Track ratio distributions. A ratio feature that changes its distribution sharply across regimes may need clipping, winsorizing, or regime-specific treatment.
Where it fits in the research system	Ratio features often feed ranking systems, filters, anomaly detectors, and liquidity checks.
Confusions to avoid	A ratio does not explain causality. It only compares magnitudes. Always ask whether numerator and denominator belong together.
Related terms	Fractions, normalization, percent change, relative strength, liquidity filters.

QUICK RECALL

Create a ratio that measures price distance from a moving average.

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Percent Algebra and Basis Points

MEANING AND MEMORY JOGGER

Meaning: Percent algebra expresses relative change per 100 units, while basis points express hundredths of one percent. Memory jogger: 1 percent = 100 basis points. 1 bp = 0.01 percent = 0.0001 as a decimal.

DEEPER EXPLANATION

Trading conversations mix decimals, percents, and basis points. Algebra keeps them consistent so cost, return, slippage, yield, and risk are not accidentally scaled wrong.

WHY IT MATTERS IN QUANT RESEARCH

A 10 bp cost is not 10 percent. Scaling mistakes of 100x or 10,000x can make a backtest look impossible or destroy a live strategy.

SIMPLE MARKET / TRADING EXAMPLE

A 25 bp fee on $100,000 is 0.0025 * 100,000 = $250.

FORMULA INTUITION

```
decimal = percent / 100
basis_points = decimal * 10000
cost_dollars = notional * bps / 10000
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Represent rates internally as decimals. Convert to percent or bps only for display and reports.
Where it fits in the research system	Cost models, yield models, risk limits, and performance reports need consistent rate units.
Confusions to avoid	Do not store some columns in percent and others in decimal without naming them explicitly, for example return_decimal vs return_pct.
Related terms	Fractions, ratios, returns, transaction costs, slippage.

QUICK RECALL

What decimal equals 35 basis points?

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Exponents and Powers

MEANING AND MEMORY JOGGER

Meaning: An exponent tells how many times a base is multiplied by itself.
Memory jogger: Exponent = repeated multiplication compressed into one symbol.

DEEPER EXPLANATION

Powers appear in compounding, squared error, variance, volatility, and nonlinear features. The exponent controls how aggressively large values are amplified.

WHY IT MATTERS IN QUANT RESEARCH

Squared returns emphasize large moves. Compound growth multiplies across periods instead of adding simple returns.

SIMPLE MARKET / TRADING EXAMPLE

A return error of 0.02 squared is 0.0004; an error of 0.10 squared is 0.0100, which is 25 times larger.

FORMULA INTUITION

```
x^2 = x * x
x^3 = x * x * x
(1 + r)^n = growth factor after n periods
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Use vectorized exponent operations carefully: df["ret_sq"] = df.ret ** 2. Check for overflow when bases or exponents are large.
Where it fits in the research system	Powers show up in risk models, loss functions, compounding engines, and nonlinear feature generation.
Confusions to avoid	Squaring removes sign. A large negative and large positive return both become large positive squared values.
Related terms	Roots, compounding, variance, loss functions, polynomial features.

QUICK RECALL

Why does squared error punish large misses more than absolute error?

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Roots and Fractional Powers

MEANING AND MEMORY JOGGER

Meaning: A root reverses a power; a fractional exponent is another way to write a root.
Memory jogger: Square root asks: what number squared gives this?

DEEPER EXPLANATION

Roots are used to undo squaring and convert variance into volatility. They also appear when annualizing volatility because independent variance scales with time and volatility scales with the square root of time.

WHY IT MATTERS IN QUANT RESEARCH

Volatility is usually the square root of variance. This makes roots a core algebra move in risk.

SIMPLE MARKET / TRADING EXAMPLE

If daily variance is 0.0001, daily volatility is sqrt(0.0001) = 0.01 = 1 percent.

FORMULA INTUITION

```
sqrt(x) = x^(1/2)
volatility = sqrt(variance)
annual_vol ≈ daily_vol * sqrt(252)
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Use np.sqrt only on nonnegative inputs. Negative variance indicates a numerical or data error.
Where it fits in the research system	Roots appear in volatility engines, risk scaling, error metrics, and transformations that compress large values.
Confusions to avoid	Do not annualize simple returns and volatility the same way. Returns often scale roughly by time; volatility scales by sqrt(time).
Related terms	Exponents, variance, volatility, annualization, numerical checks.

QUICK RECALL

Why is sqrt(252) used for volatility but not for average daily return?

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Scientific Notation and Scale Control

MEANING AND MEMORY JOGGER

Meaning: Scientific notation writes very large or very small numbers as a coefficient times a power of 10.
Memory jogger: Move the decimal, count the moves, store the scale as 10^n.

DEEPER EXPLANATION

Market systems handle tiny rates, huge notionals, microsecond timestamps, and high-volume datasets. Scientific notation keeps magnitudes readable and numerically controlled.

WHY IT MATTERS IN QUANT RESEARCH

Understanding scale prevents mistakes such as treating 1e-4 as 1 percent instead of 1 basis point.

SIMPLE MARKET / TRADING EXAMPLE

0.0001 = 1e-4. For returns, that is 1 basis point.

FORMULA INTUITION

```
a × 10^n
0.000001 = 1e-6
1,000,000 = 1e6
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Log ranges, min/max values, and dtype information during data ingestion. Float columns with extremely tiny or huge values deserve checks.
Where it fits in the research system	Scale checks protect data pipelines, cost models, optimization routines, and risk calculations from silent unit errors.
Confusions to avoid	Scientific notation changes how a number is written, not what it equals. The exponent is part of the value.
Related terms	Basis points, units, numerical stability, data validation.

QUICK RECALL

What does 2.5e-3 mean as a decimal and as a percent?

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Negative Numbers and Sign Logic

MEANING AND MEMORY JOGGER

Meaning: Sign tells direction: positive usually means gain, inflow, long exposure, or upward movement; negative usually means loss, outflow, short exposure, or downward movement.
Memory jogger: Magnitude tells size. Sign tells direction.

DEEPER EXPLANATION

Algebra with signs drives PnL, exposure, returns, deltas, and trade direction. A sign mistake can invert a signal or make a hedge increase risk instead of reducing it.

WHY IT MATTERS IN QUANT RESEARCH

Quant systems must define sign conventions explicitly. Long, short, buy, sell, debit, credit, and return direction should not be guessed.

SIMPLE MARKET / TRADING EXAMPLE

If position = -100 shares and price change = -$2, PnL = position * price_change = +$200. A short makes money when price falls.

FORMULA INTUITION

```
PnL = position_quantity * price_change
long: positive quantity
short: negative quantity
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Create tests for long and short examples. Verify a falling price produces positive PnL for a short and negative PnL for a long.
Where it fits in the research system	Sign conventions belong in the execution simulator, PnL engine, portfolio exposure model, and risk reports.
Confusions to avoid	Negative return is not always bad if the strategy is short. Always interpret return relative to exposure direction.
Related terms	PnL algebra, exposure, absolute value, inequalities, hedging.

QUICK RECALL

What happens to short PnL when price decreases?

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Linear Equations

MEANING AND MEMORY JOGGER

Meaning: A linear equation has variables only to the first power and represents a straight-line relationship.
Memory jogger: Linear means constant rate of change.

DEEPER EXPLANATION

Linear equations are the simplest algebraic models. They assume each additional unit of input changes the output by a fixed amount.

WHY IT MATTERS IN QUANT RESEARCH

Linear assumptions are useful for quick sensitivity estimates, simple forecasts, transaction-cost models, and exposure decomposition.

SIMPLE MARKET / TRADING EXAMPLE

cost = fixed_fee + spread_cost_per_share * shares. Each additional share adds the same spread cost.

FORMULA INTUITION

```
y = a + b*x
a = intercept / baseline
b = slope / sensitivity
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Fit or test linear relationships with simple regression, but inspect residuals. A line can be useful even when it is not the whole truth.
Where it fits in the research system	Linear equations are baseline models, sanity checks, and interpretable components inside larger systems.
Confusions to avoid	Linear does not mean accurate. It means the formula assumes constant marginal effect.
Related terms	Slope, intercept, regression, systems of equations, linear combinations.

QUICK RECALL

What does b mean in y = a + b*x?

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Rearranging Formulas

MEANING AND MEMORY JOGGER

Meaning: Rearranging formulas means solving for a different variable without changing the underlying relationship.
Memory jogger: Same relationship, different unknown isolated.

DEEPER EXPLANATION

A formula may be written for one purpose but needed for another. Risk formulas may be written to compute risk, but a trading system often needs to solve for size.

WHY IT MATTERS IN QUANT RESEARCH

Rearrangement is basic infrastructure for position sizing, required return, stop distance, leverage, and break-even analysis.

SIMPLE MARKET / TRADING EXAMPLE

risk = size * stop_distance. If risk budget and stop distance are known, solve size = risk / stop_distance.

FORMULA INTUITION

```
risk = size * distance
size = risk / distance
distance = risk / size
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Implement paired functions: compute_risk(size, distance) and solve_size(risk, distance). Test them against each other.
Where it fits in the research system	Formula rearrangement supports risk engines, order sizing, target setting, and scenario analysis.
Confusions to avoid	Do not rearrange across a term without preserving parentheses. risk = size * (entry - stop), not risk = size * entry - stop.
Related terms	Equality moves, solving for unknowns, break-even, PnL algebra.

QUICK RECALL

Solve notional = price * quantity for quantity.

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Solving for an Unknown

MEANING AND MEMORY JOGGER

Meaning: Solving means isolating the unknown value that makes a statement true.
Memory jogger: Move everything away from the thing you want.

DEEPER EXPLANATION

In markets, the unknown might be required price move, required win rate, implied volatility proxy, quantity, leverage, or cost threshold.

WHY IT MATTERS IN QUANT RESEARCH

A research system often computes what must happen for a trade to be worth taking, not only what already happened.

SIMPLE MARKET / TRADING EXAMPLE

If target_profit = quantity * price_move and target_profit is $500 with quantity 100, then price_move = 500 / 100 = $5.

FORMULA INTUITION

```
target_profit = qty * move
move = target_profit / qty
qty = target_profit / move
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	When solving, validate that the result is realistic and executable. A required move may be mathematically valid but impossible under current liquidity or volatility.
Where it fits in the research system	Solvers appear in sizing, calibration, threshold design, scenario testing, and risk control.
Confusions to avoid	A solved value is not a prediction. It is a requirement implied by the formula.
Related terms	Rearranging formulas, constraints, break-even, target setting.

QUICK RECALL

What unknown does your strategy need solved before placing an order?

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Inequalities and Threshold Rules

MEANING AND MEMORY JOGGER

Meaning: An inequality compares values using greater than, less than, or boundary conditions instead of exact equality.
Memory jogger: Inequality = decision boundary.

DEEPER EXPLANATION

Trading rules usually use inequalities: enter if signal > threshold, skip if spread > limit, reduce size if volatility >= cap.

WHY IT MATTERS IN QUANT RESEARCH

Inequalities turn continuous features into actions, filters, risk limits, and alerts.

SIMPLE MARKET / TRADING EXAMPLE

Enter long when price / SMA_20 > 1.02 and spread_bps < 5.

FORMULA INTUITION

```
x > threshold
x <= risk_limit
if multiply or divide by a negative number, flip the inequality sign.
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Represent rule outputs as boolean columns. Keep thresholds explicit in experiment configs, not hidden in code.
Where it fits in the research system	Inequality logic sits between feature computation and signal generation.
Confusions to avoid	Exact equality is rare with floating-point data. Use tolerances or inequality bands instead of price == level.
Related terms	Intervals, constraints, boolean masks, thresholds, decision trees.

QUICK RECALL

Why is price == moving_average usually a bad live rule?

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Intervals, Bounds, and Constraints

MEANING AND MEMORY JOGGER

Meaning: An interval describes a range of allowed values; a constraint restricts what values a system may accept.
Memory jogger: Bounds define the legal playing field.

DEEPER EXPLANATION

Most research systems need bounds: leverage must stay below a cap, weights must stay between limits, cost must stay below expected edge, parameters must remain in reasonable ranges.

WHY IT MATTERS IN QUANT RESEARCH

Constraints prevent mathematically valid but operationally absurd results.

SIMPLE MARKET / TRADING EXAMPLE

A portfolio may require 0 <= weight_i <= 0.25 and sum(weights) = 1.

FORMULA INTUITION

```
lower <= x <= upper
0 <= w_i <= max_weight
sum_i w_i = 1
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Validate constraints before simulation and again before live orders. Log rows rejected by constraint checks.
Where it fits in the research system	Constraints belong in data cleaning, feature clipping, optimization, risk, portfolio construction, and execution.
Confusions to avoid	A parameter range chosen after seeing test results can become overfitting. Choose or validate ranges honestly.
Related terms	Inequalities, portfolio weights, optimization, clipping, validation.

QUICK RECALL

Name one algebraic constraint that should exist before placing a trade.

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Absolute Value and Distance

MEANING AND MEMORY JOGGER

Meaning: Absolute value removes direction and keeps magnitude, measuring distance from zero or from a reference point.
Memory jogger: Absolute value asks: how far, ignoring sign?

DEEPER EXPLANATION

Absolute value is useful when size matters more than direction: forecast error, spread distance, deviation from average, drawdown magnitude, or exposure imbalance.

WHY IT MATTERS IN QUANT RESEARCH

Risk and diagnostics often care about magnitude. A +5 percent miss and -5 percent miss may both be equally bad for error measurement.

SIMPLE MARKET / TRADING EXAMPLE

abs(price - SMA_20) measures distance from the moving average regardless of whether price is above or below it.

FORMULA INTUITION

```
|x| = x if x >= 0, and -x if x < 0
distance = |price - reference|
absolute_error = |prediction - actual|
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Use abs() or np.abs() for vectorized distance features. Keep signed versions too if direction matters.
Where it fits in the research system	Absolute distance supports error metrics, volatility proxies, mean-reversion distance, and risk alerts.
Confusions to avoid	Absolute value destroys direction. Do not use it when long-vs-short direction matters.
Related terms	Piecewise functions, error metrics, volatility, z-score, drawdown.

QUICK RECALL

When would abs(return) be useful, and when would it hide important information?

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Piecewise Definitions

MEANING AND MEMORY JOGGER

Meaning: A piecewise function uses different formulas for different conditions.
Memory jogger: One input space, multiple rule zones.

DEEPER EXPLANATION

Markets often need conditional logic. A cost model may charge one rate for small orders and another for large orders. A signal may be long above a threshold, short below another, and flat in the middle.

WHY IT MATTERS IN QUANT RESEARCH

Piecewise definitions turn algebra into realistic trading rules, risk schedules, fee models, and regime logic.

SIMPLE MARKET / TRADING EXAMPLE

signal = 1 if z > 2, -1 if z < -2, otherwise 0.

FORMULA INTUITION

```
f(x) = { 1, if x > a
0, if -a <= x <= a
-1, if x < -a }
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Use np.select for vectorized piecewise features. Check boundary behavior explicitly: > vs >= can change signals at thresholds.
Where it fits in the research system	Piecewise logic lives in signal generation, cost models, risk tiers, regime filters, and execution rules.
Confusions to avoid	Boundary gaps and overlaps create bugs. Every possible input should map to exactly one intended branch.
Related terms	Inequalities, thresholds, decision trees, regime logic, boolean masks.

QUICK RECALL

What should happen when the input lands exactly on the threshold?

ALGEBRA / BEGINNER / FUNCTIONS

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Functions as Input-Output Machines

MEANING AND MEMORY JOGGER

Meaning: A function maps inputs to outputs according to a defined rule.
Memory jogger: Function = repeatable machine: input in, output out.

DEEPER EXPLANATION

A feature formula is a function. A risk model is a function. A backtest is a function from data and parameters to simulated results.

WHY IT MATTERS IN QUANT RESEARCH

Thinking in functions makes research modular, testable, and reproducible.

SIMPLE MARKET / TRADING EXAMPLE

f(price, SMA) = price / SMA - 1 returns percent distance from the moving average.

FORMULA INTUITION

```
y = f(x)
feature_t = f(market_data_t, parameters)
backtest_result = g(data, strategy_config)
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Write pure functions where possible: same inputs produce same outputs, no hidden state, no surprise data access.
Where it fits in the research system	Functions are building blocks for data transforms, features, models, validators, and execution simulators.
Confusions to avoid	A function is not only a math graph. In system design, any deterministic transform can be treated as a function.
Related terms	Domain, range, parameters, feature engineering, pipelines.

QUICK RECALL

What are the inputs and outputs of a moving-average feature function?

ALGEBRA / BEGINNER / FUNCTIONS

single-column field guide card

Domain and Range

MEANING AND MEMORY JOGGER

Meaning: The domain is the set of valid inputs; the range is the set of possible outputs.
Memory jogger: Domain = allowed inputs. Range = possible outputs.

DEEPER EXPLANATION

Not every formula accepts every input. Logs need positive inputs. Division needs nonzero denominators. Square roots need nonnegative inputs in real-valued systems.

WHY IT MATTERS IN QUANT RESEARCH

Ignoring domain restrictions creates NaNs, infinities, broken models, and silent row drops.

SIMPLE MARKET / TRADING EXAMPLE

log(price) is valid only when price > 0. price / volume is invalid when volume = 0.

FORMULA INTUITION

```
log(x): domain x > 0
sqrt(x): domain x >= 0
1/x: domain x != 0
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Write domain checks before transforms. Count invalid rows and decide whether to drop, impute, clip, or block the feature.
Where it fits in the research system	Domain validation belongs in the feature pipeline before modeling and backtesting.
Confusions to avoid	A formula can be algebraically valid but invalid for actual market data because the input violates its domain.
Related terms	Functions, logs, roots, denominator checks, data validation.

QUICK RECALL

What domain checks are needed before computing log(price / SMA)?

ALGEBRA / BEGINNER / FUNCTIONS

single-column field guide card

Linear Functions

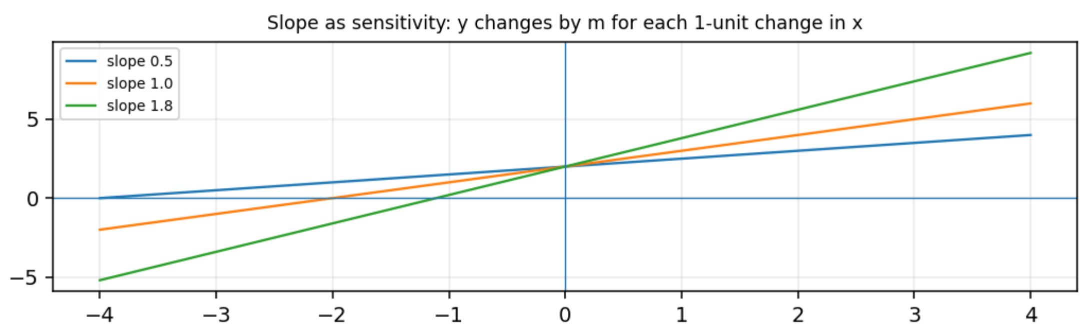
MEANING AND MEMORY JOGGER

Meaning: A linear function maps x to y using a constant slope and an intercept.
Memory jogger: Straight line, fixed slope, predictable marginal change.

DEEPER EXPLANATION

Linear functions are interpretable because each input unit has the same effect everywhere. This is why linear baselines are useful even before using complex models.

VISUAL EXAMPLE



Educational synthetic visual used for intuition; it is not a live market forecast.

WHY IT MATTERS IN QUANT RESEARCH

Linear functions support exposure estimates, simple factor models, transaction-cost approximations, and sanity checks.

SIMPLE MARKET / TRADING EXAMPLE

estimated_cost_bps = 1.5 + 0.02 * order_size_thousands.

FORMULA INTUITION

```
y = a + b*x
a = baseline when x = 0
b = change in y for each 1-unit change in x
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Plot y against x before trusting a linear assumption. If the relationship bends, add nonlinear terms or segment it piecewise.
Where it fits in the research system	Linear functions are baseline components in research pipelines and debugging dashboards.
Confusions to avoid	A line can fit the center while failing badly in tails. Market extremes often break simple linear assumptions.
Related terms	Slope, intercept, regression, sensitivity, linear combinations.

ALGEBRA / BEGINNER / FUNCTIONS

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Slope as Sensitivity

MEANING AND MEMORY JOGGER

Meaning: Slope measures how much output changes when input increases by one unit.
Memory jogger: Slope = response per unit input.

DEEPER EXPLANATION

In algebra, slope is rise over run. In research, it is sensitivity: how much cost changes with size, return changes with factor exposure, or PnL changes with price movement.

WHY IT MATTERS IN QUANT RESEARCH

Sensitivity is the language of risk. Delta, beta, factor loading, and leverage all behave like slope concepts.

SIMPLE MARKET / TRADING EXAMPLE

If cost = 2 + 0.5 * shares_in_100s, each extra 100 shares adds 0.5 cost units.

FORMULA INTUITION

```
slope = Δy / Δx
line: y = a + b*x, so b = slope
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Estimate slopes on rolling windows to see if a relationship is stable. A slope that flips sign may not be usable live.
Where it fits in the research system	Slope belongs in feature diagnostics, factor exposure, risk attribution, and model interpretation.
Confusions to avoid	Slope depends on units. Slope per dollar is different from slope per percent move.
Related terms	Linear functions, beta, delta, units, scaling.

QUICK RECALL

If x rises by 3 and y rises by 12, what is the slope?

ALGEBRA / BEGINNER / FUNCTIONS

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Intercepts and Baselines

MEANING AND MEMORY JOGGER

Meaning: An intercept is the output of a linear function when the input is zero.
Memory jogger: Intercept = starting level before x contributes.

DEEPER EXPLANATION

In trading formulas, the intercept often acts like a base cost, base return, base spread, or baseline model output.

WHY IT MATTERS IN QUANT RESEARCH

Baselines matter because a model with no input effect may still predict nonzero cost, return, or risk.

SIMPLE MARKET / TRADING EXAMPLE

cost = 1 bp + 0.02 bp per 100 shares. The 1 bp intercept is the base cost even for tiny orders.

FORMULA INTUITION

```
y = a + b*x
a = intercept = y when x = 0
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Ask whether x = 0 is meaningful. If not, center x around its average so the intercept becomes the expected output at normal conditions.
Where it fits in the research system	Intercepts appear in regressions, cost models, signal baselines, and calibration reports.
Confusions to avoid	A statistical intercept may not have a realistic market interpretation if zero input is outside observed data.
Related terms	Linear functions, slope, centering, regression, baseline forecasts.

QUICK RECALL

When is an intercept meaningful, and when is it just a mathematical artifact?

ALGEBRA / BEGINNER / SYSTEMS

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Systems of Linear Equations

MEANING AND MEMORY JOGGER

Meaning: A system of equations solves multiple relationships at the same time.
Memory jogger: Multiple equations, multiple unknowns.

DEEPER EXPLANATION

If one equation is not enough to isolate all unknowns, a system adds constraints. This is a preview of portfolio construction, hedging, and linear algebra.

WHY IT MATTERS IN QUANT RESEARCH

Hedging and allocation often require satisfying several relationships at once: total capital, sector exposure, beta exposure, or dollar neutrality.

SIMPLE MARKET / TRADING EXAMPLE

Find weights w1 and w2 such that w1 + w2 = 1 and 1.2*w1 + 0.8*w2 = 1.0 beta exposure.

FORMULA INTUITION

```
a*x + b*y = c
d*x + e*y = f
Solve both equations together, not separately.
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Use numpy.linalg.solve for small clean systems, but check condition number and constraints before trusting the answer.
Where it fits in the research system	Systems of equations connect algebra to hedging engines, optimizers, and portfolio constraint solvers.
Confusions to avoid	Two equations may have one solution, no solution, or infinitely many solutions. Do not assume a unique solution.
Related terms	Substitution, elimination, matrix solving, portfolio weights, hedging.

QUICK RECALL

Why does one equation with two unknowns usually not give one unique solution?

ALGEBRA / BEGINNER / SYSTEMS

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Substitution and Elimination

MEANING AND MEMORY JOGGER

Meaning: Substitution replaces one variable using another equation; elimination combines equations to cancel a variable.
Memory jogger: Substitution swaps. Elimination cancels.

DEEPER EXPLANATION

These are manual methods for solving systems. The logic still matters even when software solves the system automatically, because you need to understand what constraints are doing.

WHY IT MATTERS IN QUANT RESEARCH

Portfolio and hedge calculations often hide substitution and elimination inside matrix routines.

SIMPLE MARKET / TRADING EXAMPLE

If w2 = 1 - w1, substitute into beta equation to solve one unknown first.

FORMULA INTUITION

```
w1 + w2 = 1 -> w2 = 1 - w1
1.2w1 + 0.8w2 = 1.0
1.2w1 + 0.8(1 - w1) = 1.0
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	For debugging, solve a small two-asset example by hand and compare it to numpy output.
Where it fits in the research system	Substitution and elimination are useful for audit trails when an optimizer gives surprising weights.
Confusions to avoid	Do not substitute a formula derived under one constraint into a different problem with different constraints.
Related terms	Systems of equations, constraints, portfolio weights, matrix preview.

QUICK RECALL

What variable would you eliminate in a two-asset full-investment problem?

ALGEBRA / BEGINNER / SYSTEMS

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Matrix Preview for Linear Systems

MEANING AND MEMORY JOGGER

Meaning: A matrix is a rectangular structure that stores coefficients so many linear equations can be solved compactly.
Memory jogger: Matrix = coefficient table with algebra rules.

DEEPER EXPLANATION

You do not need full linear algebra yet to see the point: coefficients on variables can be stored as a table, unknowns as a vector, and targets as another vector.

WHY IT MATTERS IN QUANT RESEARCH

Factor models, regressions, covariance systems, optimizers, and portfolios all rely on matrix notation.

SIMPLE MARKET / TRADING EXAMPLE

Portfolio exposure can be written as exposures_matrix * weights = target_exposures.

FORMULA INTUITION

```
A*w = b
A = coefficient matrix
w = unknown weight vector
b = target vector
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Use matrices only after aligning rows and columns carefully. Label assets and factors so coefficient order cannot silently drift.
Where it fits in the research system	This is the bridge from beginner algebra to linear algebra, portfolio optimization, and machine learning.
Confusions to avoid	Matrix multiplication is not elementwise multiplication. Shape and alignment matter.
Related terms	Systems of equations, vectors, weights, factor exposure, linear algebra.

QUICK RECALL

What does A*w = b represent in a hedging problem?

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Quadratic Expressions

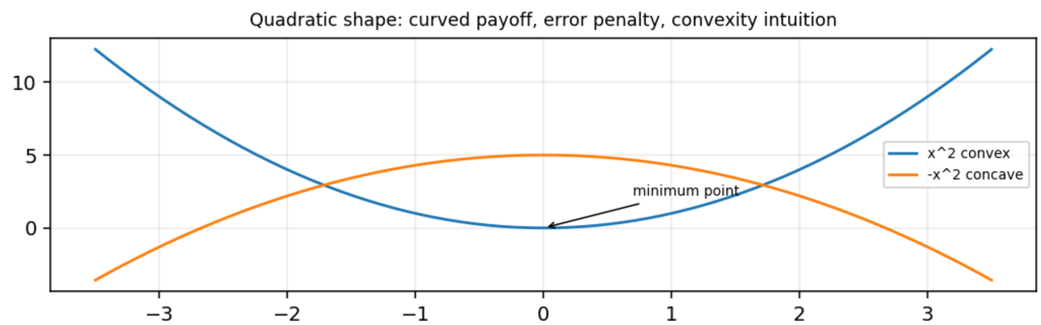
MEANING AND MEMORY JOGGER

Meaning: A quadratic expression includes a squared term such as x^2.
Memory jogger: Quadratic = curve created by squaring.

DEEPER EXPLANATION

Squared terms create curvature. They make large values grow faster than small values and often create bowl-shaped loss or risk surfaces.

VISUAL EXAMPLE



Educational synthetic visual used for intuition; it is not a live market forecast.

WHY IT MATTERS IN QUANT RESEARCH

Variance, squared error, convex risk penalties, and many optimization problems use quadratic structure.

SIMPLE MARKET / TRADING EXAMPLE

Squared forecast error = (prediction - actual)^2. A big mistake hurts disproportionately more than a small mistake.

FORMULA INTUITION

```
ax^2 + bx + c
squared_error = (y_hat - y)^2
variance uses squared deviations from the mean.
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Plot quadratic loss against error size to understand why outliers dominate squared-error metrics.
Where it fits in the research system	Quadratics appear in risk models, least-squares fitting, objective functions, and portfolio optimization.
Confusions to avoid	Squaring removes sign and amplifies magnitude. This can be good for risk, but bad if direction matters.
Related terms	Parabolas, convexity, variance, loss functions, optimization.

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Parabolas and Convexity Intuition

MEANING AND MEMORY JOGGER

Meaning: A parabola is the graph of a quadratic; convexity means the curve bends upward like a bowl.
Memory jogger: Convex bowl = one clean minimum in simple cases.

DEEPER EXPLANATION

Convex shapes are important because optimization is easier when there is a clear global minimum instead of many local traps.

WHY IT MATTERS IN QUANT RESEARCH

Portfolio risk, squared loss, and regularization often use convex penalties because they are more stable to optimize.

SIMPLE MARKET / TRADING EXAMPLE

Minimizing squared error often creates a bowl-shaped objective around the best-fitting parameter.

FORMULA INTUITION

```
f(x) = x^2 is convex
minimize f(x) when x = 0
curvature controls how penalty rises away from the minimum
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	When sweeping a parameter, plot the score curve. A smooth bowl is easier to trust than a jagged spike.
Where it fits in the research system	Convexity intuition helps diagnose optimization, parameter searches, and fragility of model selection.
Confusions to avoid	Not every objective is convex. Backtest performance surfaces are often noisy and nonconvex.
Related terms	Quadratics, loss functions, parameter heatmaps, optimization, robustness.

QUICK RECALL

Would you trust a broad stable minimum or a single sharp best point more? Why?

ALGEBRA / BEGINNER / NONLINEAR ALGEBRA

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Factoring and Common Factors

MEANING AND MEMORY JOGGER

Meaning: Factoring rewrites an expression as multiplied pieces, often revealing shared structure.
Memory jogger: Factoring pulls out what terms have in common.

DEEPER EXPLANATION

Factoring makes formulas easier to interpret, simplify, or compute safely. It can expose duplicated scaling, shared denominators, or hidden assumptions.

WHY IT MATTERS IN QUANT RESEARCH

Factoring helps simplify PnL, transaction-cost, and risk formulas so the economic drivers are clearer.

SIMPLE MARKET / TRADING EXAMPLE

price*qty - cost*qty = qty*(price - cost). Quantity is the shared factor.

FORMULA INTUITION

```
ab + ac = a(b + c)
qty*exit - qty*entry = qty*(exit - entry)
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Refactor repeated terms in code into named intermediate variables. This reduces bugs and improves auditability.
Where it fits in the research system	Factoring supports readable feature code, formula audits, and efficient calculation graphs.
Confusions to avoid	Factored and expanded forms are equivalent only if parentheses and signs are preserved exactly.
Related terms	Expressions, simplification, PnL algebra, code refactoring.

QUICK RECALL

Factor qty*exit_price - qty*entry_price.

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Completing the Square

MEANING AND MEMORY JOGGER

Meaning: Completing the square rewrites a quadratic to reveal its center and minimum or maximum point.
Memory jogger: Rewrite the curve so the turning point is visible.

DEEPER EXPLANATION

This method is less about hand calculation in trading and more about seeing that quadratic penalties have a preferred center.

WHY IT MATTERS IN QUANT RESEARCH

Squared deviations from a target are naturally written as (x - target)^2, making the best value and penalty structure clear.

SIMPLE MARKET / TRADING EXAMPLE

A risk penalty may be penalty = lambda * (exposure - target_exposure)^2.

FORMULA INTUITION

```
x^2 - 6x + 9 = (x - 3)^2
minimum occurs when x = 3
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Write penalties around target values directly: (actual - target)**2. This is clearer than expanding the terms.
Where it fits in the research system	Completing-the-square intuition appears in least squares, regularization, tracking error, and target-exposure penalties.
Confusions to avoid	Expanded quadratic forms can hide the target. Centered squared forms show the meaning immediately.
Related terms	Quadratics, convexity, squared error, objective functions.

QUICK RECALL

Why is (exposure - target)^2 easier to interpret than an expanded quadratic?

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Polynomials and Feature Expansions

MEANING AND MEMORY JOGGER

Meaning: A polynomial combines powers of a variable, such as x, x^2, and x^3.
Memory jogger: Polynomial = sum of powered terms.

DEEPER EXPLANATION

Polynomial features let a model capture curvature while still using algebraic building blocks. They can also overfit quickly if added blindly.

WHY IT MATTERS IN QUANT RESEARCH

Feature expansions can model nonlinear relationships such as volatility effects, saturation, or accelerating costs.

SIMPLE MARKET / TRADING EXAMPLE

cost = a + b*size + c*size^2 allows large orders to become disproportionately more expensive.

FORMULA INTUITION

```
p(x) = a0 + a1*x + a2*x^2 + ... + an*x^n
quadratic cost = a + bq + cq^2
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Scale variables before polynomial expansion. Raw large values raised to powers can dominate models and create numerical instability.
Where it fits in the research system	Polynomial features belong in model research, cost modeling, and nonlinear signal experiments.
Confusions to avoid	More powers do not automatically mean better signal. They may just memorize noise.
Related terms	Exponents, quadratics, feature engineering, overfitting, scaling.

QUICK RECALL

Why might size^2 be useful in a transaction-cost model?

ALGEBRA / BEGINNER / GROWTH AND SCALE

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Exponential Growth and Decay

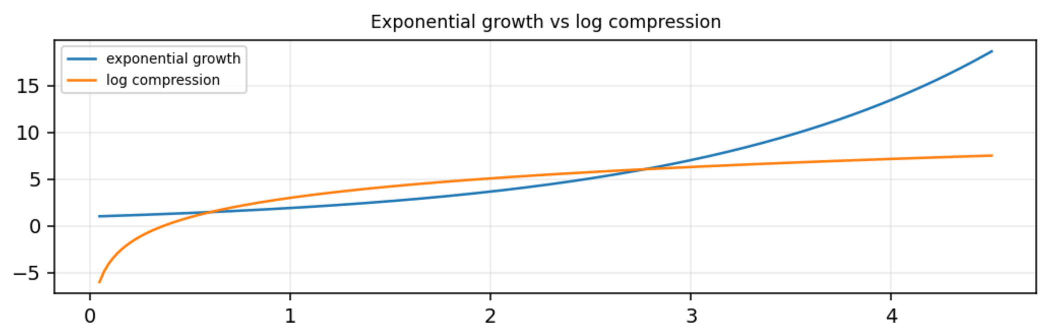
MEANING AND MEMORY JOGGER

Meaning: Exponential formulas multiply by a constant growth or decay factor over steps.
Memory jogger: Exponential means the base keeps compounding.

DEEPER EXPLANATION

Compounding is exponential because each period acts on the new level, not only the starting level. Decay works the same way in reverse.

VISUAL EXAMPLE



Educational synthetic visual used for intuition; it is not a live market forecast.

WHY IT MATTERS IN QUANT RESEARCH

Capital growth, drawdown recovery, half-life, discounting, and exponential moving averages all use exponential thinking.

SIMPLE MARKET / TRADING EXAMPLE

A 1 percent gain compounded for 10 periods is (1.01)^10 - 1, not simply 10 percent exactly.

FORMULA INTUITION

```
future = present * (1 + r)^n
decay = initial * λ^n, where 0 < λ < 1
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Use log1p for small returns when accumulating compounded returns: np.log1p(r).cumsum().
Where it fits in the research system	Exponential formulas appear in performance compounding, smoothing, decay weights, and risk memory.
Confusions to avoid	Adding returns is an approximation. Compounding multiplies growth factors.
Related terms	Logs, geometric sequences, EMA, compounding, half-life.

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Logarithms as Inverse Growth

MEANING AND MEMORY JOGGER

Meaning: A logarithm answers what exponent is needed to produce a value from a base.
Memory jogger: Log asks: what power got me here?

DEEPER EXPLANATION

Logs compress large scales, turn multiplication into addition, and make compounded growth easier to work with.

WHY IT MATTERS IN QUANT RESEARCH

Log transforms help with price relatives, cumulative returns, multiplicative processes, skewed variables, and numerical stability.

SIMPLE MARKET / TRADING EXAMPLE

If wealth grows by factors 1.02 and 0.99, log growth adds: log(1.02) + log(0.99).

FORMULA INTUITION

```
if b^y = x, then log_b(x) = y
log(a*b) = log(a) + log(b)
log(a/b) = log(a) - log(b)
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Check positivity before logs. Use np.log1p(r) for returns near zero because it is more stable than np.log(1+r).
Where it fits in the research system	Logs appear in return pipelines, scale compression, transformed features, and cumulative performance engines.
Confusions to avoid	Log input must be positive. A zero or negative price-relative breaks the transform.
Related terms	Exponents, log returns, compounding, domain checks, scaling.

QUICK RECALL

Why do logs make compounded growth easier to aggregate?

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Natural Log Rules for Finance

MEANING AND MEMORY JOGGER

Meaning: The natural log uses base e and is the standard log for continuous compounding and many finance formulas.
Memory jogger: ln is the finance default because compounding math becomes clean.

DEEPER EXPLANATION

Natural logs show up in log returns, option pricing, growth rates, and continuous-time approximations. The main beginner idea: ln turns price ratios into additive growth increments.

WHY IT MATTERS IN QUANT RESEARCH

Log returns are additive across time, which makes them useful in research pipelines and performance aggregation.

SIMPLE MARKET / TRADING EXAMPLE

ln(P_t / P_{t-1}) is the one-period log return.

FORMULA INTUITION

```
ln(a*b) = ln(a) + ln(b)
ln(a/b) = ln(a) - ln(b)
log_return_t = ln(P_t / P_{t-1})
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Use np.log(price).diff() or np.log(price / price.shift(1)). Validate price > 0 first.
Where it fits in the research system	Natural logs connect algebra foundations to returns, volatility, time series, and portfolio analytics.
Confusions to avoid	ln is a specific log base. In Python, np.log means natural log, not base-10 log.
Related terms	Logarithms, log returns, compounding, time series, domain checks.

QUICK RECALL

What does np.log compute in Python?

ALGEBRA / BEGINNER / SEQUENCES AND SUMS

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Arithmetic Sequences

MEANING AND MEMORY JOGGER

Meaning: An arithmetic sequence changes by adding the same amount each step.
Memory jogger: Arithmetic sequence = constant difference.

DEEPER EXPLANATION

Arithmetic sequences describe evenly spaced grids, fixed step schedules, linear ramps, and simple parameter sweeps.

WHY IT MATTERS IN QUANT RESEARCH

Parameter sweeps, threshold grids, and walk-forward window starts often use arithmetic stepping.

SIMPLE MARKET / TRADING EXAMPLE

Test thresholds 0.5, 1.0, 1.5, 2.0. Each step adds 0.5.

FORMULA INTUITION

```
a_n = a_1 + (n - 1)d
d = common difference
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Use np.arange or np.linspace intentionally. Know whether the endpoint is included.
Where it fits in the research system	Arithmetic sequences help define research grids, time windows, and deterministic schedules.
Confusions to avoid	A fixed additive step is different from a fixed percentage step. Choose based on what scale matters.
Related terms	Geometric sequences, parameter grids, timeline diagrams, loops.

QUICK RECALL

When would a parameter grid use additive steps instead of multiplicative steps?

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Geometric Sequences

MEANING AND MEMORY JOGGER

Meaning: A geometric sequence changes by multiplying by the same factor each step.
Memory jogger: Geometric sequence = constant ratio.

DEEPER EXPLANATION

Geometric sequences model compounding, exponential decay, and multiplicative parameter grids.

WHY IT MATTERS IN QUANT RESEARCH

Lookbacks, risk multipliers, growth factors, and decay weights often make more sense on a multiplicative scale.

SIMPLE MARKET / TRADING EXAMPLE

Lookbacks 5, 10, 20, 40, 80 double each step.

FORMULA INTUITION

```
a_n = a_1 * r^(n - 1)
r = common ratio
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Use log-spaced grids for parameters that operate by scale, such as regularization strength or lookback length.
Where it fits in the research system	Geometric sequences support compounding engines, decay-weighted features, and robust parameter searches.
Confusions to avoid	A geometric sequence can grow fast. Avoid huge parameter ranges without reason.
Related terms	Exponents, exponential growth, compounding, parameter grids.

QUICK RECALL

Why might lookbacks use 5, 10, 20, 40 instead of 5, 10, 15, 20?

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Sigma Notation

MEANING AND MEMORY JOGGER

Meaning: Sigma notation is a compact way to write a sum over many terms.
Memory jogger: Sigma means add the indexed pieces.

DEEPER EXPLANATION

A lot of quant math is repeated addition: averages, cumulative returns, total PnL, squared errors, portfolio totals, and rolling window calculations.

WHY IT MATTERS IN QUANT RESEARCH

Sigma notation is the bridge between formulas and vectorized code.

SIMPLE MARKET / TRADING EXAMPLE

The average of returns is the sum of returns divided by the number of returns.

FORMULA INTUITION

```
sum_{i=1}^{n} x_i = x_1 + x_2 + ... + x_n
mean = (1/n) * sum_{i=1}^{n} x_i
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Vectorized code: x.sum(), x.mean(), df.groupby(...).sum(). Make sure the index set matches the intended window.
Where it fits in the research system	Sums appear in indicators, loss functions, portfolio aggregation, and performance reports.
Confusions to avoid	The index bounds matter. Rolling sums, expanding sums, and full-sample sums answer different questions.
Related terms	Averages, rolling windows, loss functions, portfolio totals.

QUICK RECALL

What does the i index represent in a rolling 20-day sum?

ALGEBRA / BEGINNER / SEQUENCES AND SUMS

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Product Notation

MEANING AND MEMORY JOGGER

Meaning: Product notation is a compact way to write repeated multiplication.
Memory jogger: Product symbol means multiply the indexed pieces.

DEEPER EXPLANATION

Compounded returns multiply growth factors. Product notation makes the difference between additive and multiplicative accumulation explicit.

WHY IT MATTERS IN QUANT RESEARCH

Cumulative wealth and compounded performance depend on products, not sums, when returns are applied sequentially.

SIMPLE MARKET / TRADING EXAMPLE

If returns are 2 percent and -1 percent, wealth factor = 1.02 * 0.99 = 1.0098.

FORMULA INTUITION

```
prod_{i=1}^{n} x_i = x_1 * x_2 * ... * x_n
wealth_factor = prod_t (1 + r_t)
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Use (1 + returns).cumprod() for equity curves. Use log sums for more stable long-horizon compounding.
Where it fits in the research system	Product notation underlies equity curves, portfolio growth, and multiplicative probability-style calculations.
Confusions to avoid	Summing returns can approximate short periods, but it is not the exact compounded wealth path.
Related terms	Geometric sequences, exponential growth, log returns, equity curves.

QUICK RECALL

Why does an equity curve use cumulative product?

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Recursion and Iterative Updates

MEANING AND MEMORY JOGGER

Meaning: A recursive formula defines the next value using previous values.
Memory jogger: Next state depends on prior state.

DEEPER EXPLANATION

Many market calculations update through time: equity, drawdown, exponential averages, portfolio holdings, cash balance, and regime state.

WHY IT MATTERS IN QUANT RESEARCH

Backtests are recursive because today's state depends on yesterday's position, cash, equity, and orders.

SIMPLE MARKET / TRADING EXAMPLE

equity_t = equity_{t-1} * (1 + strategy_return_t).

FORMULA INTUITION

```
x_t = f(x_{t-1}, input_t)
equity_t = equity_{t-1} + PnL_t
EMA_t = α*price_t + (1-α)*EMA_{t-1}
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Avoid look-ahead by ensuring updates use only past state and current allowed data. Unit-test the first few rows manually.
Where it fits in the research system	Recursion is central to backtesting, stateful indicators, execution simulation, and risk tracking.
Confusions to avoid	Vectorized code can hide recursion. Confirm that future values are not leaking into current state.
Related terms	Sequences, EMA, backtesting, state machines, look-ahead bias.

QUICK RECALL

What state variables does a backtest carry from one bar to the next?

ALGEBRA / BEGINNER / PORTFOLIO ALGEBRA

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Weighted Averages

MEANING AND MEMORY JOGGER

Meaning: A weighted average multiplies each value by an assigned weight and adds the results.
Memory jogger: Weighted average = values times importance.

DEEPER EXPLANATION

Equal averages treat every input the same. Weighted averages let larger positions, newer observations, or more reliable sources matter more.

WHY IT MATTERS IN QUANT RESEARCH

Portfolio return, index return, VWAP, factor exposure, and ensemble forecasts are weighted averages.

SIMPLE MARKET / TRADING EXAMPLE

Portfolio return = 0.60 * stock_return + 0.40 * bond_return.

FORMULA INTUITION

```
weighted_avg = sum_i w_i*x_i
usually sum_i w_i = 1
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Check weight sums and alignment. A correct formula with misaligned assets creates wrong portfolio returns.
Where it fits in the research system	Weighted averages are everywhere in portfolio construction, aggregation, smoothing, and model ensembles.
Confusions to avoid	Weights must match the values they multiply. Asset order mistakes are silent and dangerous.
Related terms	Linear combinations, portfolio weights, sums, VWAP, EMA.

QUICK RECALL

Why should weights usually add to 1?

single-column field guide card

Linear Combinations

MEANING AND MEMORY JOGGER

Meaning: A linear combination adds scaled versions of variables.
Memory jogger: Scale each piece, then add them.

DEEPER EXPLANATION

Weighted averages are special linear combinations. More generally, linear combinations can represent portfolios, factor models, signals, and hedges.

WHY IT MATTERS IN QUANT RESEARCH

Portfolio PnL is a linear combination of position PnLs. A factor model is a linear combination of factor exposures.

SIMPLE MARKET / TRADING EXAMPLE

signal = 0.5*momentum + 0.3*value - 0.2*volatility.

FORMULA INTUITION

```
y = a1*x1 + a2*x2 + ... + an*xn
portfolio_return = sum_i w_i*r_i
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Use vector dot products for linear combinations: np.dot(weights, returns). Confirm vector order and labels.
Where it fits in the research system	Linear combinations sit in signal blending, portfolio construction, regression, hedging, and factor models.
Confusions to avoid	Adding features with different scales can make the largest-scale feature dominate accidentally.
Related terms	Weighted averages, matrix preview, portfolio returns, factor models.

QUICK RECALL

What happens if one feature is measured in dollars and another in decimals before combining them?

ALGEBRA / BEGINNER / FEATURE ENGINEERING

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Normalization by Algebra

MEANING AND MEMORY JOGGER

Meaning: Normalization rescales raw values so they can be compared or modeled more fairly.
Memory jogger: Normalize means put values onto a useful common scale.

DEEPER EXPLANATION

Raw market variables have different units and magnitudes. Algebraic normalization converts them into comparable features.

WHY IT MATTERS IN QUANT RESEARCH

Models and rules often behave better when features have stable, interpretable scales.

SIMPLE MARKET / TRADING EXAMPLE

price_distance = (price - moving_average) / moving_average normalizes distance by the average price level.

FORMULA INTUITION

```
relative_distance = (x - reference) / reference
scaled_value = x / baseline
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Choose the baseline carefully: current price, rolling average, volume average, volatility, or capital. The baseline defines the meaning.
Where it fits in the research system	Normalization sits between raw data and feature stores. It makes cross-asset and cross-period comparisons possible.
Confusions to avoid	Normalization can remove useful level information. Keep raw and normalized versions when testing.
Related terms	Ratios, fractions, z-scores, min-max scaling, feature stores.

QUICK RECALL

What baseline would you use to normalize volume?

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Min-Max Scaling

MEANING AND MEMORY JOGGER

Meaning: Min-max scaling maps values into a fixed range, often 0 to 1.
Memory jogger: Subtract the floor, divide by the range.

DEEPER EXPLANATION

Min-max scaling is simple and intuitive, but it is sensitive to outliers and regime changes because the min and max define the whole scale.

WHY IT MATTERS IN QUANT RESEARCH

It can help with bounded indicators, visualization, and models requiring comparable feature ranges.

SIMPLE MARKET / TRADING EXAMPLE

If RSI-like feature ranges from 20 to 80, a value of 50 maps to (50 - 20) / (80 - 20) = 0.5.

FORMULA INTUITION

```
x_scaled = (x - min_x) / (max_x - min_x)
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Fit min and max on training data only, then apply to validation/test data. Never use future test extrema to scale past features.
Where it fits in the research system	Min-max scaling belongs in the feature pipeline with saved train-set scaling metadata.
Confusions to avoid	A new live value above the training max can produce a scaled value above 1. That is not automatically a bug; it may be regime change.
Related terms	Normalization, z-score, leakage, feature scaling, outliers.

QUICK RECALL

Why is fitting min/max on the entire dataset a leakage problem?

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Z-Score Standardization

MEANING AND MEMORY JOGGER

Meaning: A z-score measures how many standard deviations a value is from its mean.
Memory jogger: Z-score = distance from average in volatility units.

DEEPER EXPLANATION

Standardization subtracts a mean and divides by a standard deviation. It turns raw deviations into a common distance scale.

WHY IT MATTERS IN QUANT RESEARCH

Z-scores are common in mean reversion, anomaly detection, ranking, normalization, and model inputs.

SIMPLE MARKET / TRADING EXAMPLE

If price spread is 3 and its mean is 1 with standard deviation 2, $z = (3 - 1) / 2 = 1$.

FORMULA INTUITION

```
z = (x - μ) / σ
μ = mean
σ = standard deviation
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Use rolling or training-window μ and σ. Watch for σ near zero; it causes unstable z-scores.
Where it fits in the research system	Z-score standardization sits in feature stores, anomaly filters, and signal thresholds.
Confusions to avoid	A z-score assumes the chosen mean and standard deviation are relevant. In regime shifts, old μ and σ can mislead.
Related terms	Mean, standard deviation, normalization, volatility, regime shifts.

QUICK RECALL

What does $z = -2$ mean in plain English?

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Break-Even Algebra

MEANING AND MEMORY JOGGER

Meaning: Break-even algebra solves the minimum outcome needed to offset costs or losses.
Memory jogger: Break-even asks: what value makes net result zero?

DEEPER EXPLANATION

A trade can have a positive raw move and still lose after spread, commission, slippage, borrow, funding, or taxes. Break-even formulas expose the hurdle.

WHY IT MATTERS IN QUANT RESEARCH

Every strategy needs to beat its costs. Algebra makes that requirement explicit before modeling edge.

SIMPLE MARKET / TRADING EXAMPLE

If round-trip cost is 8 bps, the expected gross move must exceed 8 bps before the trade has positive net edge.

FORMULA INTUITION

```
net_return = gross_return - cost
break_even when net_return = 0
gross_required = cost
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Build cost-adjusted features and cost-adjusted backtests from the start. Report gross and net results separately.
Where it fits in the research system	Break-even logic belongs in signal filters, cost models, backtests, and deployment thresholds.
Confusions to avoid	A high win rate can still lose if average win is too small after cost.
Related terms	Cost model, expected value, PnL algebra, thresholds, slippage.

QUICK RECALL

What cost has to be beaten before a strategy has edge?

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Position Sizing Algebra

MEANING AND MEMORY JOGGER

Meaning: Position sizing algebra converts risk budget, stop distance, price, and contract multiplier into trade size.
Memory jogger: Size comes from risk allowed divided by risk per unit.

DEEPER EXPLANATION

Sizing should not be guessed. If you know how much you are willing to lose and how much one unit can lose, the formula gives maximum units.

WHY IT MATTERS IN QUANT RESEARCH

Sizing connects signals to risk. A good signal with bad sizing can still destroy a portfolio.

SIMPLE MARKET / TRADING EXAMPLE

Risk budget = $200. Stop distance = $2 per share. Size = 200 / 2 = 100 shares.

FORMULA INTUITION

```
units = risk_budget / risk_per_unit
risk_per_unit = |entry - stop| * multiplier
notional = units * price * multiplier
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Round size down to valid lot size. Recompute risk after rounding and enforce max notional/leverage constraints.
Where it fits in the research system	Sizing logic belongs between signal generation and order simulation/execution.
Confusions to avoid	Sizing by conviction alone ignores volatility and stop distance. Two trades with equal conviction can need different sizes.
Related terms	Absolute value, PnL algebra, constraints, leverage, risk budget.

QUICK RECALL

Why does wider stop distance reduce allowable size?

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PnL Algebra

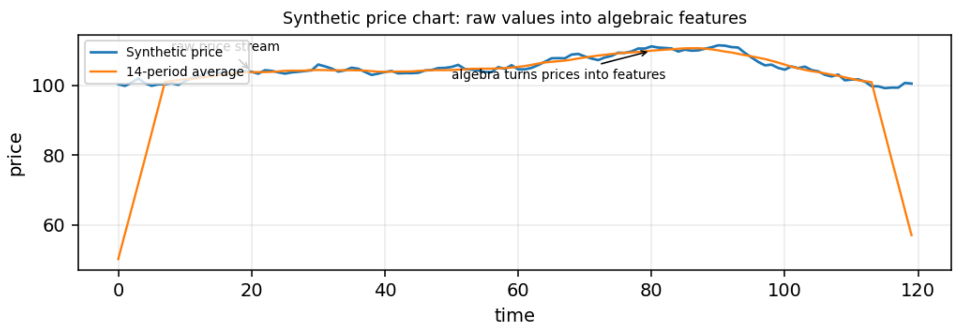
MEANING AND MEMORY JOGGER

Meaning: PnL algebra computes profit or loss from position, price movement, multiplier, and costs.
Memory jogger: PnL = exposure times movement minus costs.

DEEPER EXPLANATION

The exact formula depends on instrument type, but the algebraic structure is consistent: quantity, direction, price change, multiplier, and costs.

VISUAL EXAMPLE



Educational synthetic visual used for intuition; it is not a live market forecast.

WHY IT MATTERS IN QUANT RESEARCH

Backtests are only as good as their PnL accounting. A sign, multiplier, or cost error can invalidate the entire result.

SIMPLE MARKET / TRADING EXAMPLE

Long 50 shares from \$100 to \$103 with \$2 total cost: PnL = 50*(103 - 100) - 2 = \$148.

FORMULA INTUITION

```
long_pnl = qty*(exit - entry)*multiplier - costs
short_pnl = qty*(entry - exit)*multiplier - costs
signed_pnl = signed_qty*(exit - entry)*multiplier - costs
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Create fixtures for long winner, long loser, short winner, short loser, zero move, and cost-only cases.
Where it fits in the research system	PnL algebra is the core of backtesting, live reconciliation, attribution, and risk reporting.
Confusions to avoid	Contract multipliers and sign conventions are common sources of false backtest results.
Related terms	Sign logic, position sizing, costs, multipliers, backtesting.

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Portfolio Weight Constraints

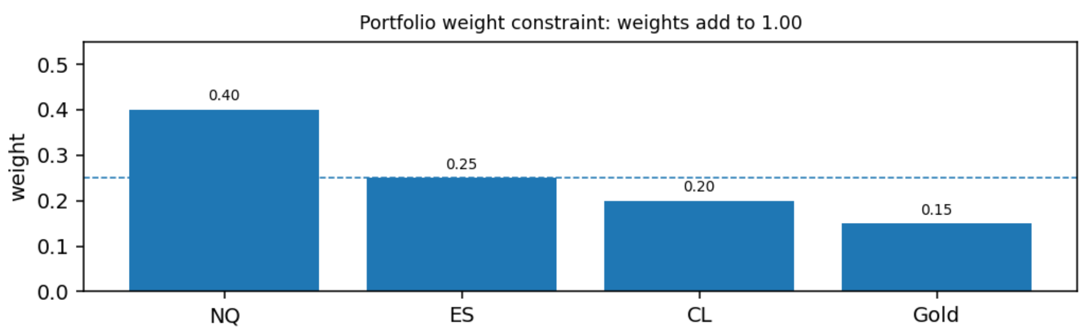
MEANING AND MEMORY JOGGER

Meaning: Portfolio weights describe what fraction of capital is assigned to each asset or strategy.
Memory jogger: Weights are allocation fractions.

DEEPER EXPLANATION

Weights turn individual returns into portfolio returns. Constraints define whether the portfolio is long-only, leveraged, market-neutral, capped, or fully invested.

VISUAL EXAMPLE



Educational synthetic visual used for intuition; it is not a live market forecast.

WHY IT MATTERS IN QUANT RESEARCH

Portfolio algebra is mostly weight algebra: sum constraints, exposure constraints, and risk constraints.

SIMPLE MARKET / TRADING EXAMPLE

If weights are 0.5, 0.3, and 0.2, they sum to 1.0 and represent 100 percent of capital.

FORMULA INTUITION

```
portfolio_return = sum_i w_i*r_i
fully invested: sum_i w_i = 1
long-only: w_i >= 0
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Check weight sums at every rebalance. Store weights by timestamp and asset ID so they can be joined to returns safely.
Where it fits in the research system	Weight constraints sit in portfolio construction, optimizer outputs, backtest accounting, and risk dashboards.
Confusions to avoid	Weights can be negative for shorts and can sum above 1 under leverage. Define the convention before interpreting results.
Related terms	Weighted averages, linear combinations, constraints, exposure, leverage.

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Indicator Algebra: SMA and EMA

MEANING AND MEMORY JOGGER

Meaning: Moving averages are algebraic summaries of past values; SMA weights recent observations equally, while EMA decays older observations.
Memory jogger: SMA = equal memory. EMA = fading memory.

DEEPER EXPLANATION

Indicators are not magic. They are formulas that transform raw price or volume into smoother features.

WHY IT MATTERS IN QUANT RESEARCH

Understanding indicator algebra prevents treating indicators as black boxes. It also helps avoid look-ahead and alignment errors.

SIMPLE MARKET / TRADING EXAMPLE

SMA_20 at time t should use the last 20 available prices, not future prices.

FORMULA INTUITION

```
SMA_n(t) = (1/n) * sum_{i=0}^{n-1} P_{t-i}
EMA_t = αP_t + (1-α)EMA_{t-1}
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Shift indicators if the strategy can only act after the bar closes. Confirm whether your library includes current bar data.
Where it fits in the research system	Indicator algebra lives in the feature layer and must be timestamp-aligned before signal generation.
Confusions to avoid	A moving average is a transform, not a strategy. The strategy starts when you define rules using it.
Related terms	Sigma notation, recursion, EMA, feature engineering, look-ahead bias.

QUICK RECALL

Why might an unshifted moving average create look-ahead in a backtest?

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Feature Engineering as Formula Design

MEANING AND MEMORY JOGGER

Meaning: Feature engineering designs algebraic transforms that turn raw data into model-ready or rule-ready inputs.
Memory jogger: Feature = raw data transformed into a useful measurement.

DEEPER EXPLANATION

Good features have clear meaning, stable construction, valid timestamps, and sensible scale. Algebra is the first language of feature design.

WHY IT MATTERS IN QUANT RESEARCH

Most quant research is not about inventing complex models first. It is about creating useful measurements and testing whether they hold up.

SIMPLE MARKET / TRADING EXAMPLE

momentum_20 = price / price.shift(20) - 1 measures 20-period relative change.

FORMULA INTUITION

```
feature_t = transform(raw_data_{<=t}, parameters)
momentum_n = P_t/P_{t-n} - 1
vol_adjusted_momentum = momentum_n / volatility_n
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Document every feature with inputs, timestamp rules, parameters, domain checks, and expected range.
Where it fits in the research system	Feature engineering is the bridge from data ingestion to model/rule evaluation.
Confusions to avoid	A feature with future data leakage can look predictive while being impossible to trade.
Related terms	Functions, ratios, normalization, rolling windows, leakage, feature store.

QUICK RECALL

What raw data and timestamp assumptions does your feature use?

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Parameters vs Variables in Research

MEANING AND MEMORY JOGGER

Meaning: Variables are data values; parameters are adjustable settings chosen by the researcher or model.
Memory jogger: Data changes by market. Parameters change by experiment.

DEEPER EXPLANATION

A variable like return_t comes from the market. A parameter like lookback = 20 comes from the research design. Confusing them makes validation messy.

WHY IT MATTERS IN QUANT RESEARCH

Parameter discipline is central to avoiding overfitting. Every parameter choice needs a reason, a search space, or a validation plan.

SIMPLE MARKET / TRADING EXAMPLE

In zscore_n = (x_t - rolling_mean_n) / rolling_std_n, x_t is a variable and n is a parameter.

FORMULA INTUITION

```
output_t = f(data_t, parameter_set)
parameter_set = {lookback: 20, threshold: 2.0}
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Hash or log parameter sets for every backtest. Reproducing results requires exact parameters, not just code.
Where it fits in the research system	Experiment tracking depends on clean separation of data variables and research parameters.
Confusions to avoid	Optimized parameters are not facts about the market. They are choices that may or may not generalize.
Related terms	Variables, feature engineering, overfitting, validation, experiment tracking.

QUICK RECALL

What parameters would you log for a moving-average crossover test?

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Objective Functions and Loss Algebra

MEANING AND MEMORY JOGGER

Meaning: An objective function scores how good or bad a choice is so a system can compare alternatives.
Memory jogger: Objective = what the system is trying to optimize.

DEEPER EXPLANATION

A loss function is an objective you minimize. A performance score is often an objective you maximize. Algebra defines exactly what success means.

WHY IT MATTERS IN QUANT RESEARCH

Changing the objective changes the strategy. Maximizing raw return is different from maximizing risk-adjusted return or minimizing drawdown.

SIMPLE MARKET / TRADING EXAMPLE

Mean squared error penalizes forecast errors; net Sharpe-like scores penalize volatile returns and costs.

FORMULA INTUITION

```
MSE = (1/n) * sum_i (prediction_i - actual_i)^2
score = expected_return - cost_penalty - risk_penalty
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Write objective functions explicitly and test edge cases. Check whether the objective rewards behavior you actually want.
Where it fits in the research system	Objectives guide model training, parameter search, portfolio optimization, and strategy selection.
Confusions to avoid	A high objective score can be gamed if the objective omits costs, liquidity, turnover, or tail risk.
Related terms	Quadratics, squared error, optimization, validation, overfitting.

QUICK RECALL

What behavior does your objective reward and what does it ignore?

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Units, Ticks, and Multipliers

MEANING AND MEMORY JOGGER

Meaning: Units define what a number measures; ticks and multipliers translate price movement into money.
Memory jogger: Same price move, different contract, different dollar effect.

DEEPER EXPLANATION

Many trading mistakes are unit mistakes. A futures tick, stock penny, forex pip, and crypto dollar move are not interchangeable.

WHY IT MATTERS IN QUANT RESEARCH

Backtests need instrument specifications. Without tick size and multiplier, PnL and risk can be wrong even if direction is correct.

SIMPLE MARKET / TRADING EXAMPLE

If multiplier is $50 per index point, a 2-point move equals $100 per contract before costs.

FORMULA INTUITION

```
dollar_move = price_move * multiplier * contracts
ticks = price_move / tick_size
tick_value = tick_size * multiplier
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Create an instrument metadata table: symbol, tick_size, multiplier, currency, trading hours, fee model, lot size.
Where it fits in the research system	Units and multipliers belong in the instrument master and are consumed by backtests, risk, and execution code.
Confusions to avoid	Do not assume one price point equals one dollar. The multiplier defines dollar exposure.
Related terms	Dimensional analysis, PnL algebra, position sizing, instrument metadata.

QUICK RECALL

What metadata is required to turn futures price movement into dollars?

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Dimensional Analysis

MEANING AND MEMORY JOGGER

Meaning: Dimensional analysis checks whether the units on both sides of a formula make sense.
Memory jogger: The units should balance like the algebra.

DEEPER EXPLANATION

If a formula returns dollars, the ingredients should combine to dollars. If it returns a percent, it should be dimensionless. Unit mismatches catch many hidden bugs.

WHY IT MATTERS IN QUANT RESEARCH

Dimensional checks prevent mixing returns, dollars, shares, contracts, basis points, and prices incorrectly.

SIMPLE MARKET / TRADING EXAMPLE

shares * dollars_per_share = dollars. But shares + dollars is meaningless.

FORMULA INTUITION

```
quantity [shares] * price_change [$ / share] = PnL [$]
return [unitless] = price_change [$] / price [$]
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Use explicit column names: price_usd, return_decimal, cost_bps, quantity_shares. Names act as lightweight unit checks.
Where it fits in the research system	Dimensional analysis protects data engineering, feature engineering, PnL, risk, and reporting.
Confusions to avoid	A spreadsheet can add incompatible units without error. Your system should not.
Related terms	Units, ratios, PnL algebra, returns, metadata, validation.

QUICK RECALL

What units should result from quantity * price_change?

ALGEBRA / BEGINNER / SYSTEM DESIGN

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Numerical Stability and Algebraic Forms

MEANING AND MEMORY JOGGER

Meaning: Numerical stability means choosing algebraic forms that behave well under finite-precision computation.
Memory jogger: Equivalent on paper does not always mean equally safe in code.

DEEPER EXPLANATION

Computers use finite precision. Very small differences, huge numbers, division by tiny values, and subtracting nearly equal values can create unstable results.

WHY IT MATTERS IN QUANT RESEARCH

Research pipelines can produce false signals when numerical artifacts look like market structure.

SIMPLE MARKET / TRADING EXAMPLE

Computing log(1+r) with tiny r is less stable than using log1p(r).

FORMULA INTUITION

```
prefer: np.log1p(r) for log(1+r) when r is small
avoid: division by near-zero denominators
check: abs(denominator) > epsilon
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Add finite checks: np.isfinite, denominator eps guards, clipping rules, and warnings for extreme values.
Where it fits in the research system	Numerical stability belongs in feature pipelines, optimizers, risk engines, and production monitors.
Confusions to avoid	A massive feature value may be a denominator bug, not a market signal.
Related terms	Domain checks, ratios, logs, scaling, validation, data quality.

QUICK RECALL

Why can mathematically equivalent formulas behave differently in Python?

ALGEBRA / BEGINNER / SYSTEM DESIGN

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Algebra Debugging in Research Code

MEANING AND MEMORY JOGGER

Meaning: Algebra debugging checks whether formulas, signs, units, and timestamps match the intended research logic.
Memory jogger: Debug the math before judging the strategy.

DEEPER EXPLANATION

A backtest can be wrong because of one formula: unshifted feature, inverted sign, missing multiplier, wrong denominator, or parameter leakage.

WHY IT MATTERS IN QUANT RESEARCH

Most beginner strategy failures are not subtle math; they are basic algebra and data-alignment mistakes.

SIMPLE MARKET / TRADING EXAMPLE

A strategy that buys when price is below average but reports trend-following behavior may have an inverted comparison or sign.

FORMULA INTUITION

```
debug checklist:
1. units balance
2. signs match direction
3. denominators valid
4. timestamps use only known data
5. parameters logged
```

IMPLEMENTATION, SYSTEM FIT, AND TRAPS

Module	Reference note
Python / system implementation	Create tiny toy datasets where the correct output is obvious. Run every formula against them before full backtests.
Where it fits in the research system	Formula debugging should be a required gate before model training, backtesting, and live simulation.
Confusions to avoid	Good-looking results do not validate formula correctness. They may be exploiting a bug.
Related terms	Unit tests, PnL algebra, timestamp alignment, leakage, validation.

QUICK RECALL

What is the smallest toy example that proves your PnL formula works?

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Formula Summary Sheet

CORE FORMULAS

Concept	Formula	Use
Relative change	(new - old) / old	Convert raw move into comparable scale.
Linear equation	y = a + b*x	Baseline and constant sensitivity.
Slope	Δy / Δx	Sensitivity per input unit.
Z-score	(x - μ) / σ	Distance from baseline in volatility units.
Weighted average	sum_i w_i*x_i	Portfolio return, exposure, VWAP.
Cumulative wealth	prod_t (1 + r_t)	Exact compounded return path.
PnL	signed_qty*(exit-entry)*multiplier - costs	Backtest accounting.
Position size	risk_budget / risk_per_unit	Risk-controlled sizing.
Min-max scaling	(x - min) / (max - min)	Map values into 0-1 range.
EMA recursion	αx_t + (1-α)EMA_{t-1}	Decay-weighted smoothing.

FORMULA AUDIT CHECKLIST

For every formula, write: input columns, timestamp rule, denominator check, output unit, parameter values, expected range, and one toy example with a known answer.

MINIMUM SAFE FEATURE DEFINITION

```
feature_name: vol_adjusted_momentum_20
inputs: P_t, P_{t-20}, rolling_vol_20
formula: (P_t/P_{t-20} - 1) / rolling_vol_20
domain: P_t > 0, P_{t-20} > 0, rolling_vol_20 > epsilon
timestamp: available only after bar t closes
```

QUICK RECALL

Quick recall: a formula is research-ready only when it is mathematically defined, timestamp-safe, unit-checked, parameter-logged, and tested on a toy example.

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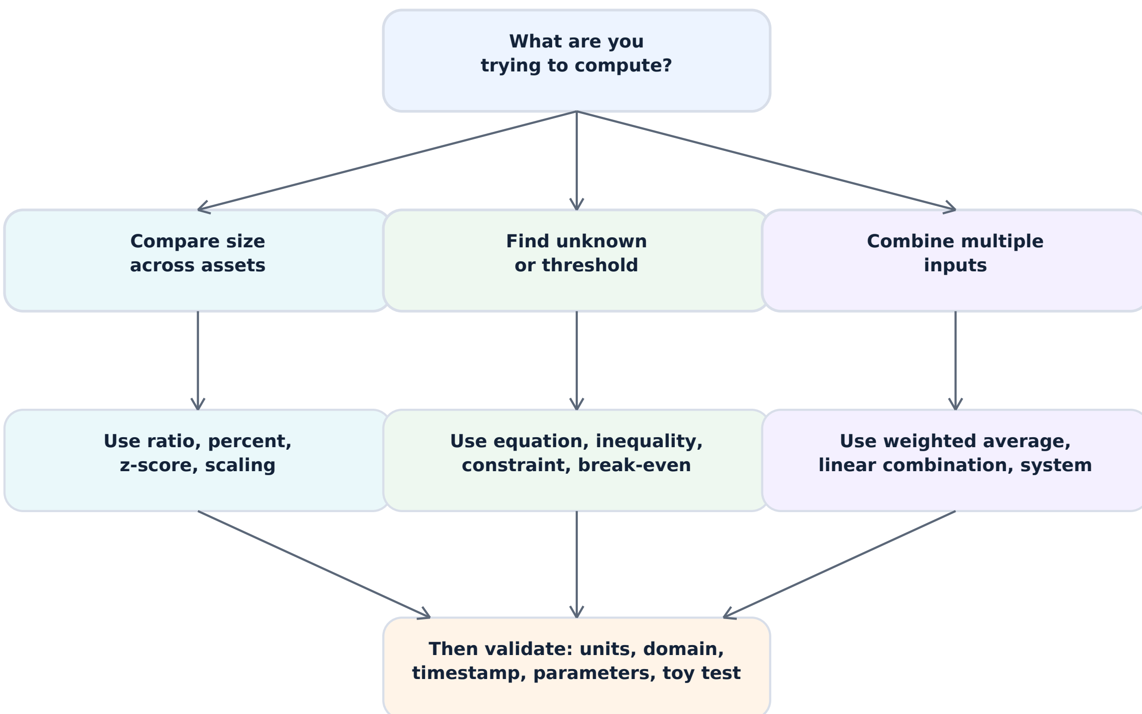
Algebra Method Selection Decision Tree

PURPOSE

Use this decision tree when choosing the algebraic form for a feature, rule, sizing formula, or diagnostic.

FULL-WIDTH DECISION TREE



METHOD SELECTION SHORTCUTS

Need	Likely algebra
Relative comparison	Ratio, percent change, normalization
Decision boundary	Inequality, threshold, interval
Risk-controlled size	Rearranged equation
Portfolio aggregation	Weighted average / linear combination
State through time	Recursion
Outlier distance	Absolute value or z-score

ALGEBRA / BEGINNER / REVIEW

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Common Failure Modes and Fixes

DIAGNOSTIC TABLE

Failure mode	Symptom	Likely cause	Fix
Wrong denominator	Huge or unstable ratios	Zero or tiny base value	Epsilon guard, NaN filter, baseline review
Sign inversion	Long logic behaves short	Position or comparison sign wrong	Toy long/short tests
Unit mismatch	PnL off by 10x or 100x	Ticks, bps, percent, multiplier mixed	Instrument metadata + unit names
Look-ahead	Backtest too smooth or good	Feature uses future data	Shift features, timestamp audit
Parameter leakage	Test results degrade live	Parameters chosen on test data	Train/validation split, experiment logs
Boundary bug	Missing trades at threshold	> vs >= mismatch	Explicit boundary tests
Numerical instability	Inf, NaN, extreme features	Division, log domain, overflow	Domain checks, clipping, stable functions

MINIMUM DEBUGGING WORKFLOW

Start with a 5-row toy dataset. Compute the formula by hand. Run the code. Compare every intermediate column. Only then run the full backtest. If the toy case fails, the full backtest is not evidence.

ALGEBRA TEST FIXTURE PATTERN

```
input: entry=100, exit=103, qty=50, multiplier=1, costs=2
expected long pnl = 50*(103-100)-2 = 148
expected short pnl = 50*(100-103)-2 = -152
```

QUICK RECALL

Recall: most formula bugs are not advanced math failures. They are denominator, sign, unit, timestamp, or parameter-control failures.

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Research Pipeline Algebra Checkpoints

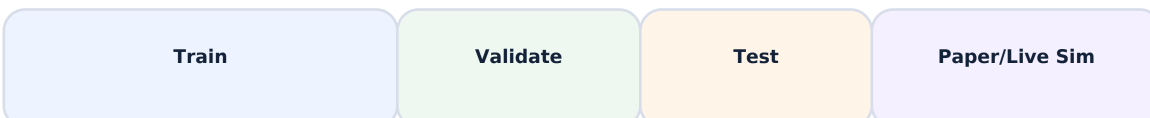
PIPELINE PRINCIPLE

Each stage should preserve algebraic meaning: the same units, signs, timestamps, and parameter definitions should survive from raw data through live simulation.

PIPELINE CHECKPOINT TABLE

Stage	Input	Algebra check	Output
Ingest	Raw prices, volume, metadata	Units, tick size, multiplier, missing values	Clean aligned data
Transform	Clean data	Domain, denominator, rolling-window alignment	Feature table
Signal	Features	Thresholds, inequalities, piecewise branch coverage	Action candidates
Sizing	Signal + risk budget	Position formula, constraints, rounding	Order quantities
Backtest	Orders + prices	PnL, costs, recursion, timestamps	Equity curve
Validate	Results + logs	Parameter stability, failure modes, sensitivity	Go/no-go decision

WALK-FORWARD TIMELINE: PARAMETER DISCIPLINE



Choose or fit parameters in training. Use validation to select. Use test and paper/live simulation to estimate generalization, not to keep tuning forever.

QUICK RECALL

Checkpoint question: which algebraic decisions are allowed to learn from which period?

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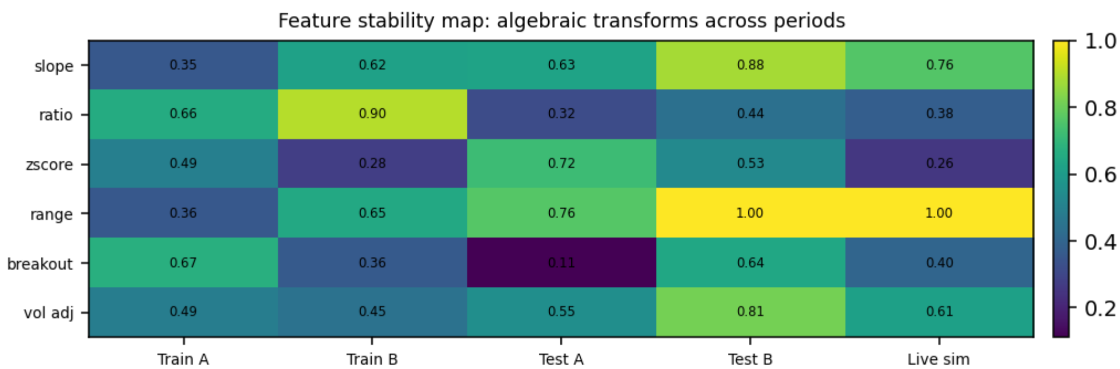
Feature Stability and Parameter Sensitivity

PURPOSE

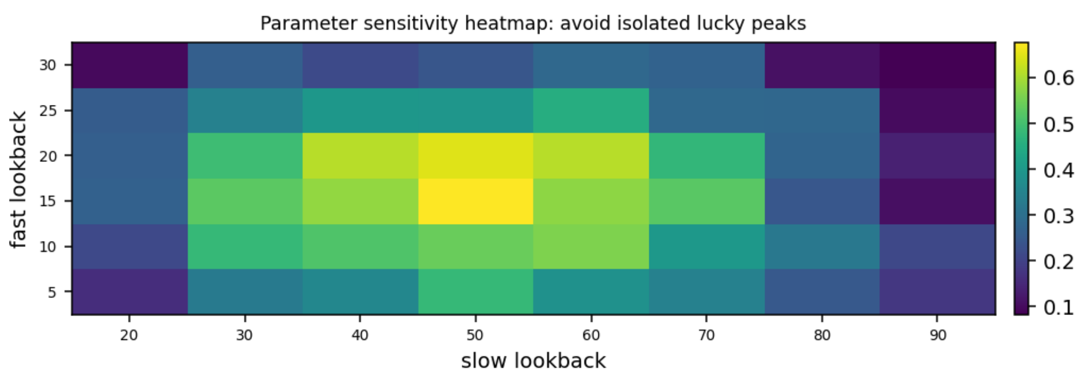
Visual diagnostics reveal whether algebraic features and parameter choices are stable across time, regimes, and search ranges. Stability matters more than a single lucky best result.

FEATURE STABILITY MAP



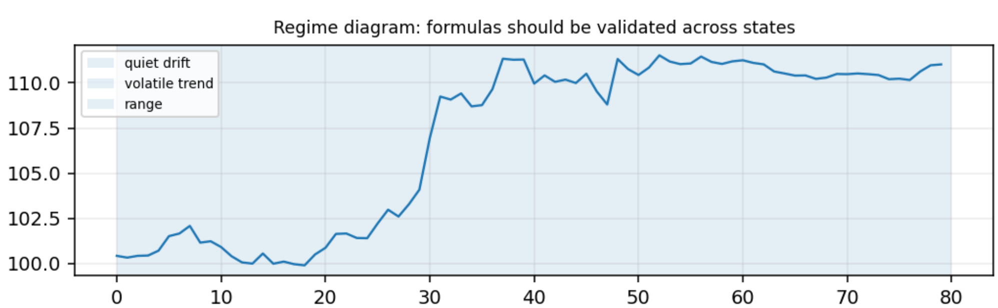
Look for features that remain useful across train, test, and live-like periods instead of only one isolated period.

PARAMETER SENSITIVITY HEATMAP



A broad plateau is usually more robust than a single sharp peak. Sharp peaks can be a sign of overfit parameter selection.

REGIME TIMELINE



The same algebraic feature can behave differently in quiet, trending, volatile, or range-bound regimes.

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Quick Recall Review and Index

QUICK RECALL PROMPTS

Prompt	Answer pattern
How do I compare across assets?	Use ratios, percent change, normalization, or z-scores.
How do I turn a feature into a rule?	Use inequality thresholds or piecewise logic.
How do I solve for size?	Rearrange risk = size * risk_per_unit.
How do I combine assets or features?	Use weighted averages or linear combinations.
How do I check a formula?	Units, signs, domain, timestamp, parameters, toy test.
How do I avoid leakage?	Fit parameters and scalers only on allowed training data.

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FINAL MENTAL MODEL

Algebra is not separate from quant research. It is the operating grammar of features, rules, constraints, sizing, PnL, risk, and validation. When a strategy fails, inspect the algebra before blaming the market or the model.

QUICK RECALL

Review drill: choose any strategy idea and write the variables, parameters, formulas, constraints, PnL equation, and first toy test.